TT1, EEE 459 **Marks: 20** **Time: 30 minutes**

Term Test 1 (its questions 1 to 3 are 7(a) to 7(c))

[TT1, EEE 459; Marks: 20; Time: 30 minutes]

7. (a) Show the compositional dependence of direct-energy gap Γ and indirect energy gap X for GaPAs. Which one is more stepper [*sic*: should be steeper] and why? 1+2

   (b) Explain how the density of states changes qualitatively as the dimensionality reduces from 3D → 2D → 1D → 0D. Sketch $D(E)$ vs E for each case. 8

   (c) A metal-semiconductor (n-type) junction is formed where the work function of metal is greater than that of the semiconductor. With necessary assumptions, derive expression for electric field in both regions. Also sketch the electric field profile for the both regions. 9

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